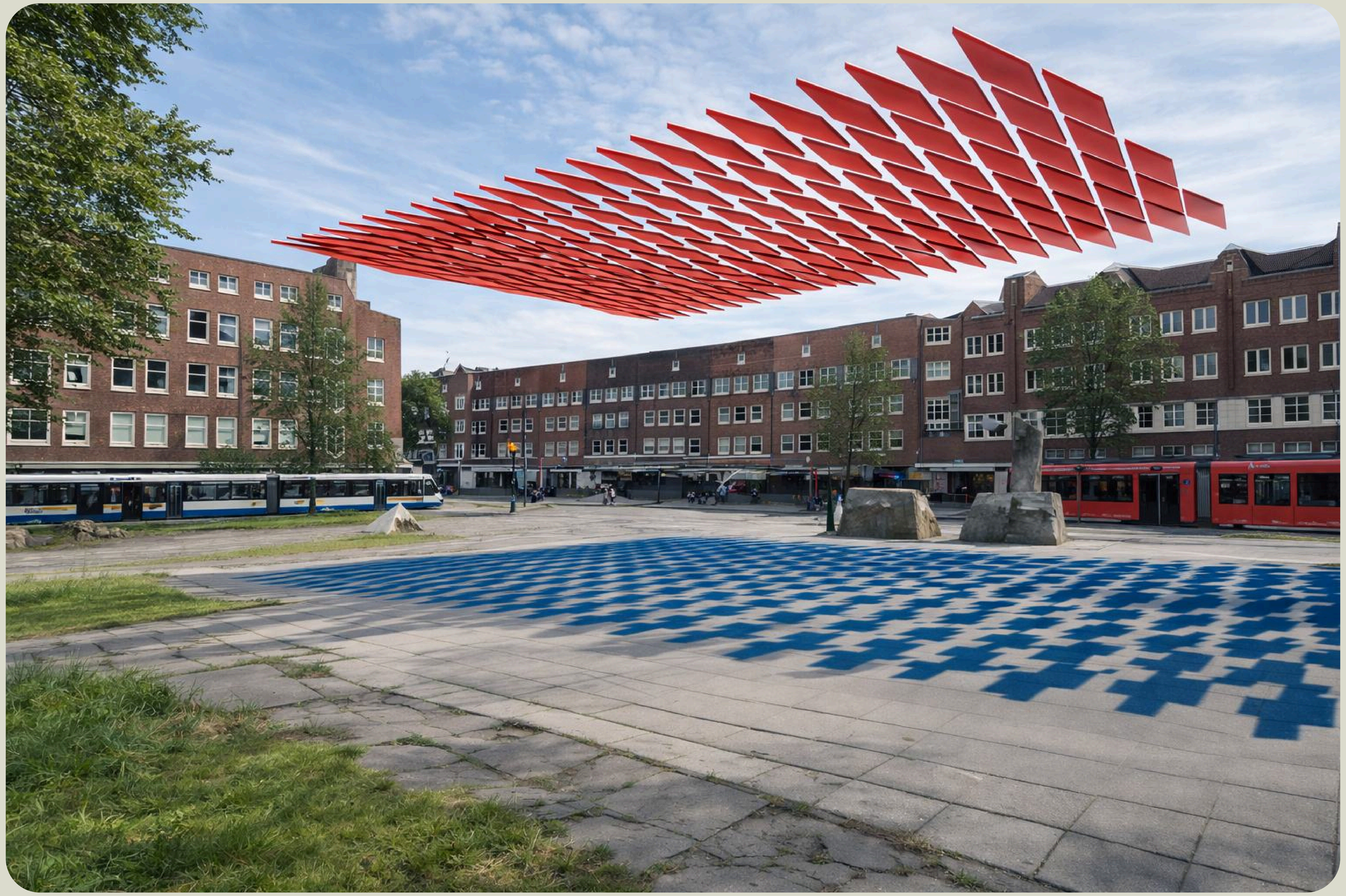
AI Generated Shade

Designing a parametric shading canopy to mitigate the Urban Heat Island effect



AI ENHANCED RENDITION

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COMPUTATIONAL INTELLIGENCE FOR INTEGRATED DESIGN

15/04/2026

Agenda

- Background
- Methodology
- ML Algorithms
- Presenting Choices
- Implications

The Urban Heat Island Effect

- Urban areas experience higher temperatures than rural areas
- This phenomenon is known as the Urban Heat Island effect
- Mainly caused by human activities and infrastructure
- Mitigating this effect is crucial for sustainable urban living

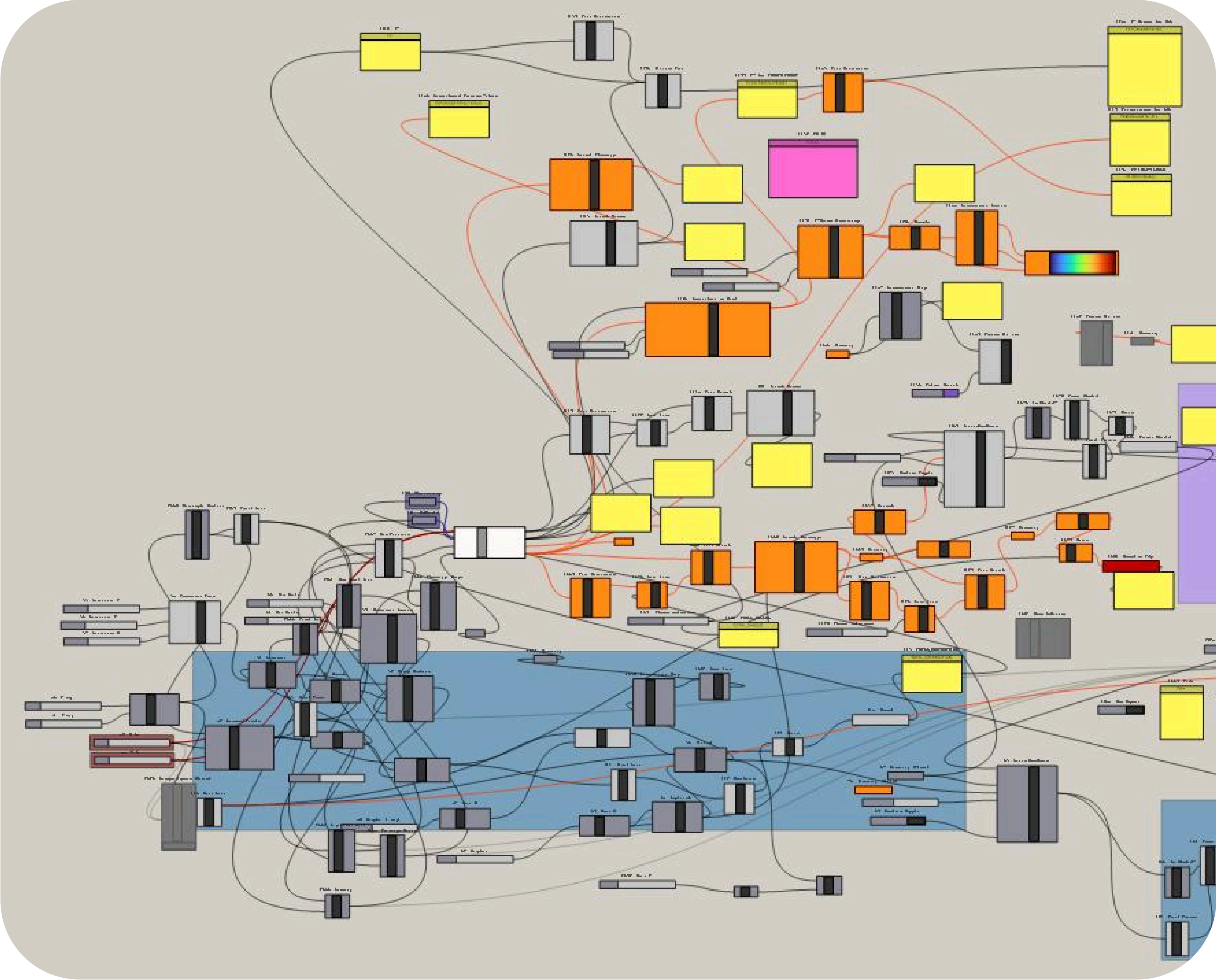
Design Assignment

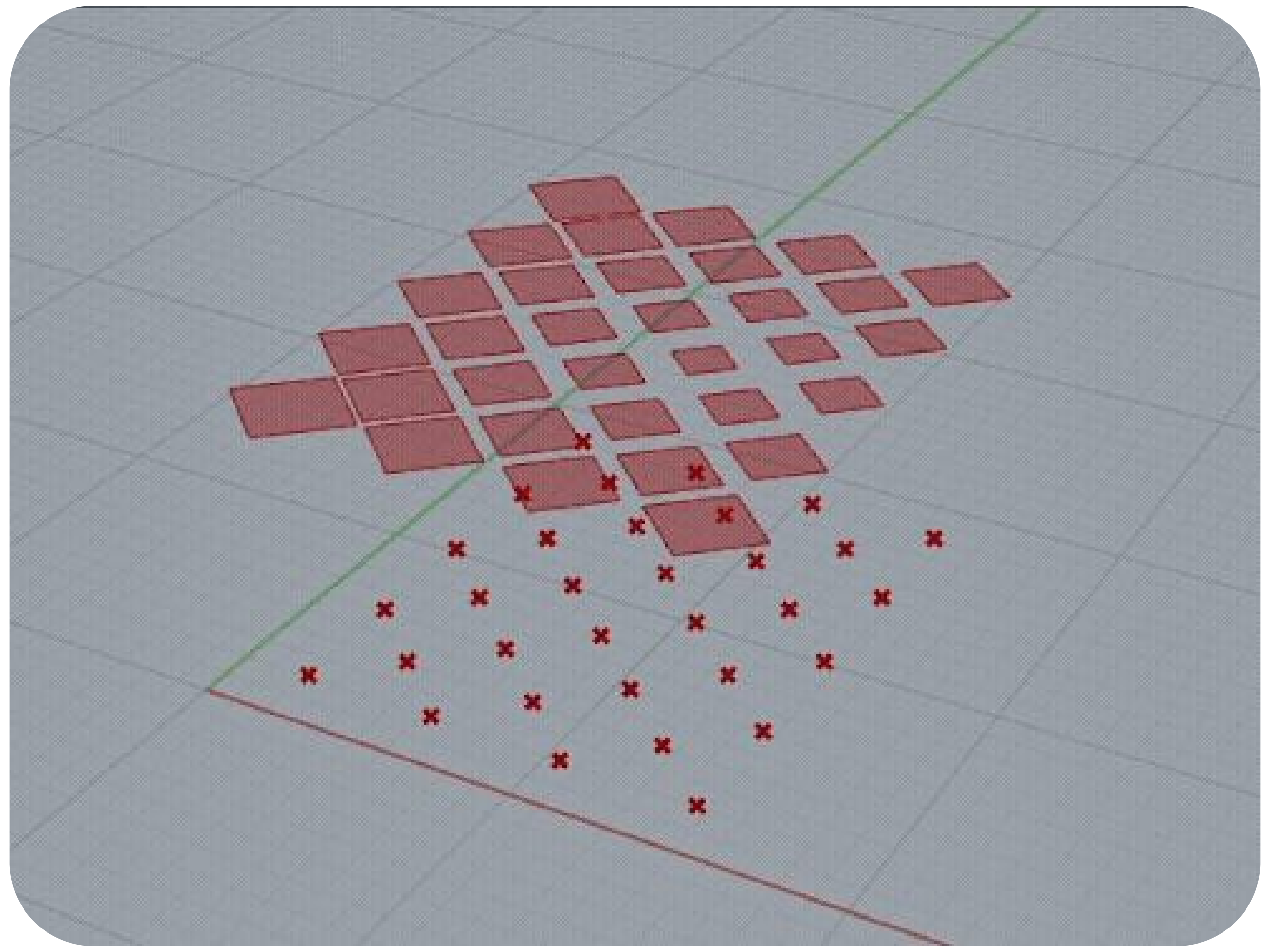
- Create a solution that helps mitigate the urban heat island effect
- Explore how computational intelligence can support design and decision making
- Apply 2 ML algorithms
- Present results effectively
- Explore how AI can support human capabilities

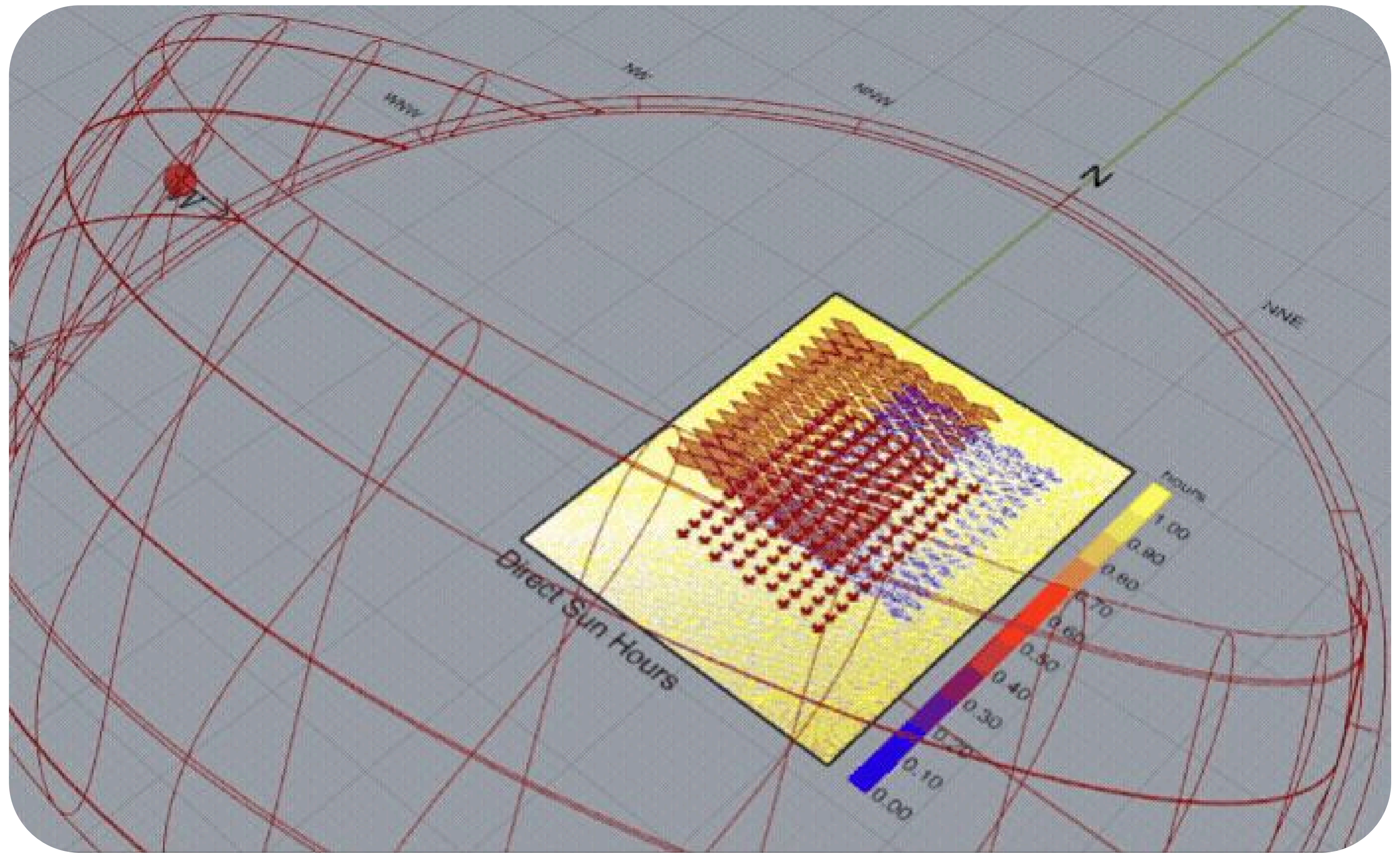
Exploration and Approach

- Inspiration
- Creating a parametric shading canopy
- Diamond grid by altering U and V divisions
- Goal: Maximize Shade, Minimize Sky Obstruction
- Choosing a location
- Sunlight Analysis for Amsterdam
- Analysis for hottest day of the year

THE HAGUE CENTRAL STATION, BENTHEM CROUWEL



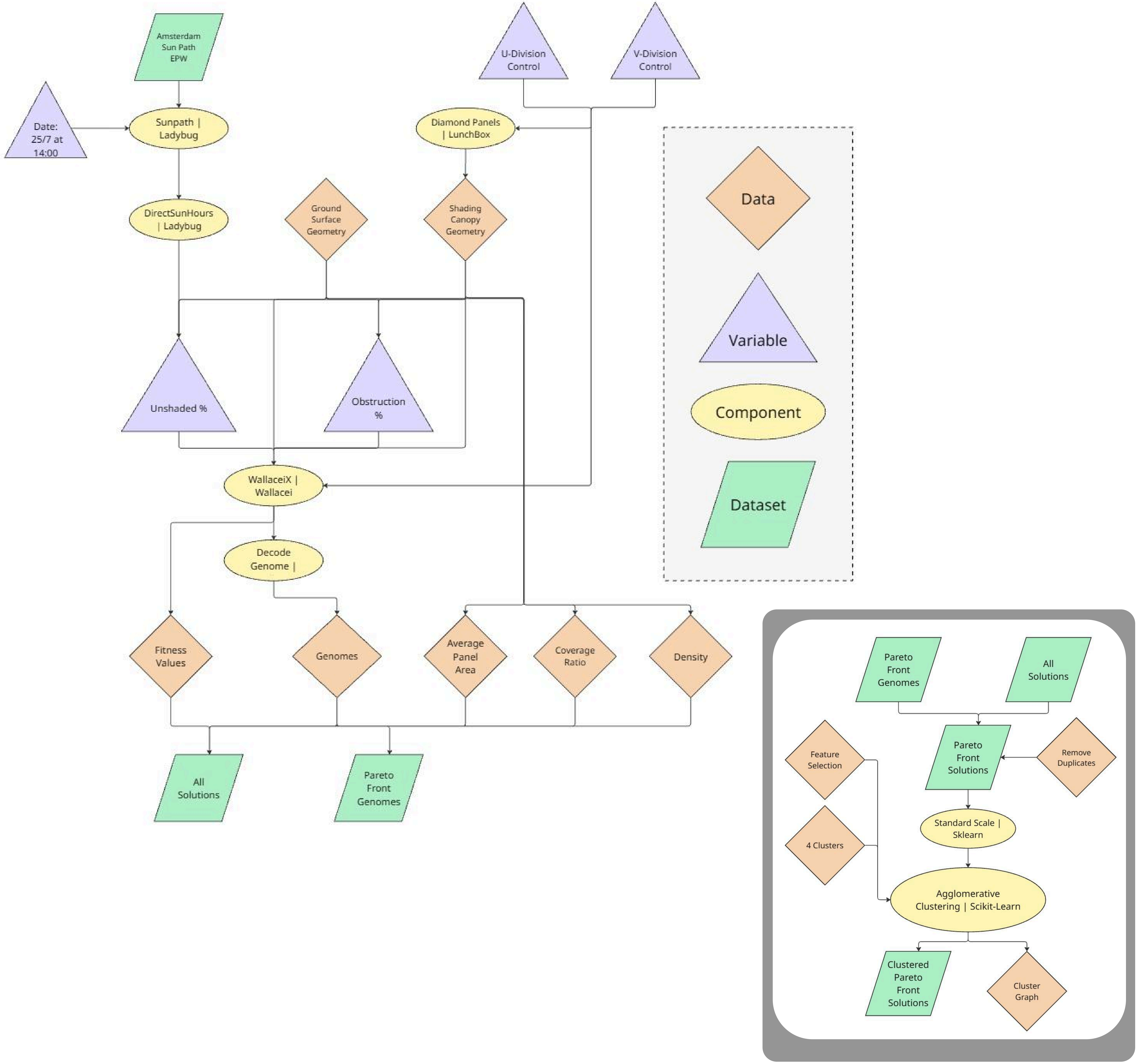




Pipeline

Generation and Classification

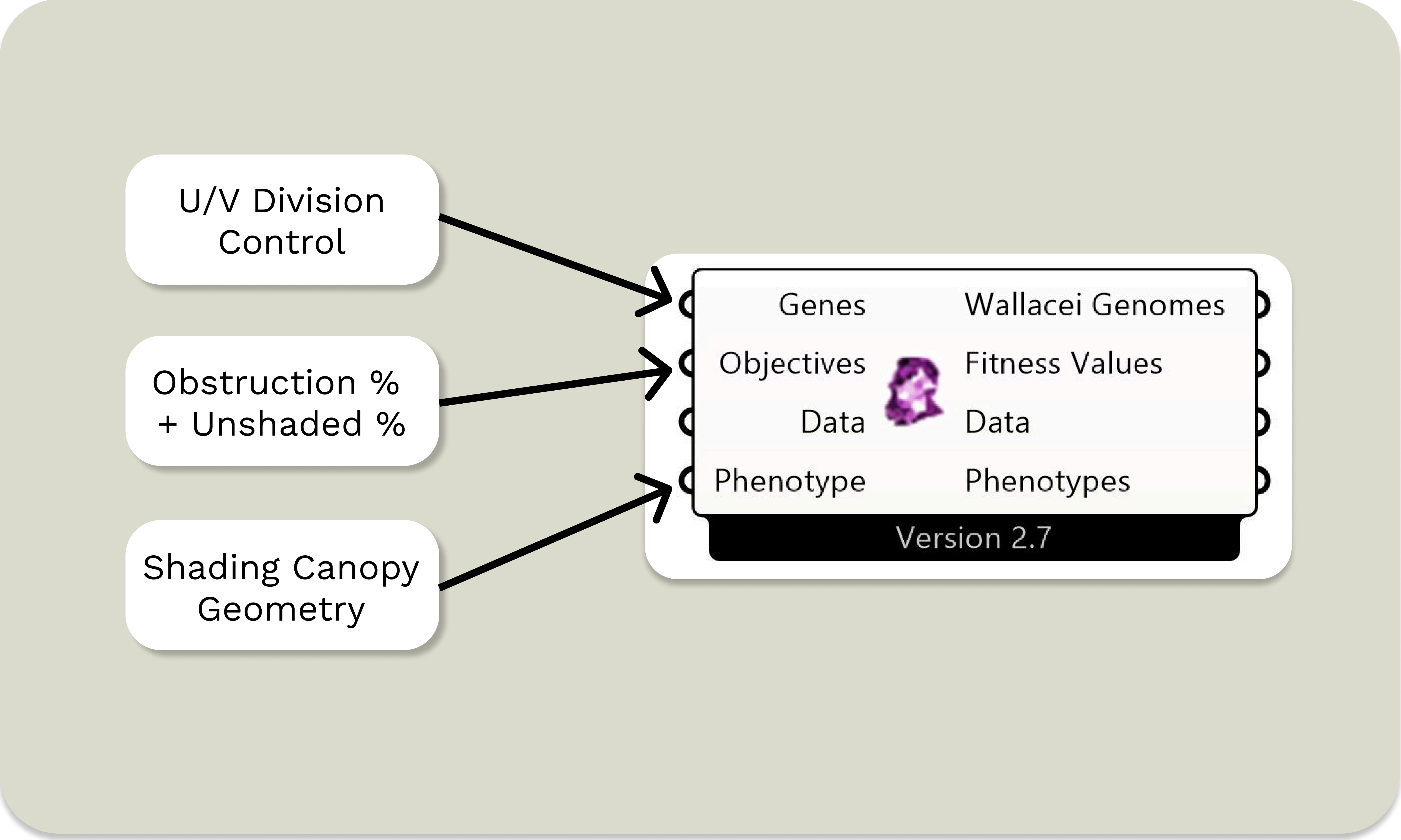
- Software
  - Rhino 8
  - Grasshopper
  - VScode
- Datasets
  - EPW for Amsterdam sun path
  - Generated Pareto Front solutions
- Plugins
  - LadyBug
  - LunchBox
  - Wallacei
  - Raven
- ML Algorithms
  - WallaceiX Evolutionary Algorithm
  - Agglomerative Clustering - Scikit



Evolutionary Algorithm

WallaceiX

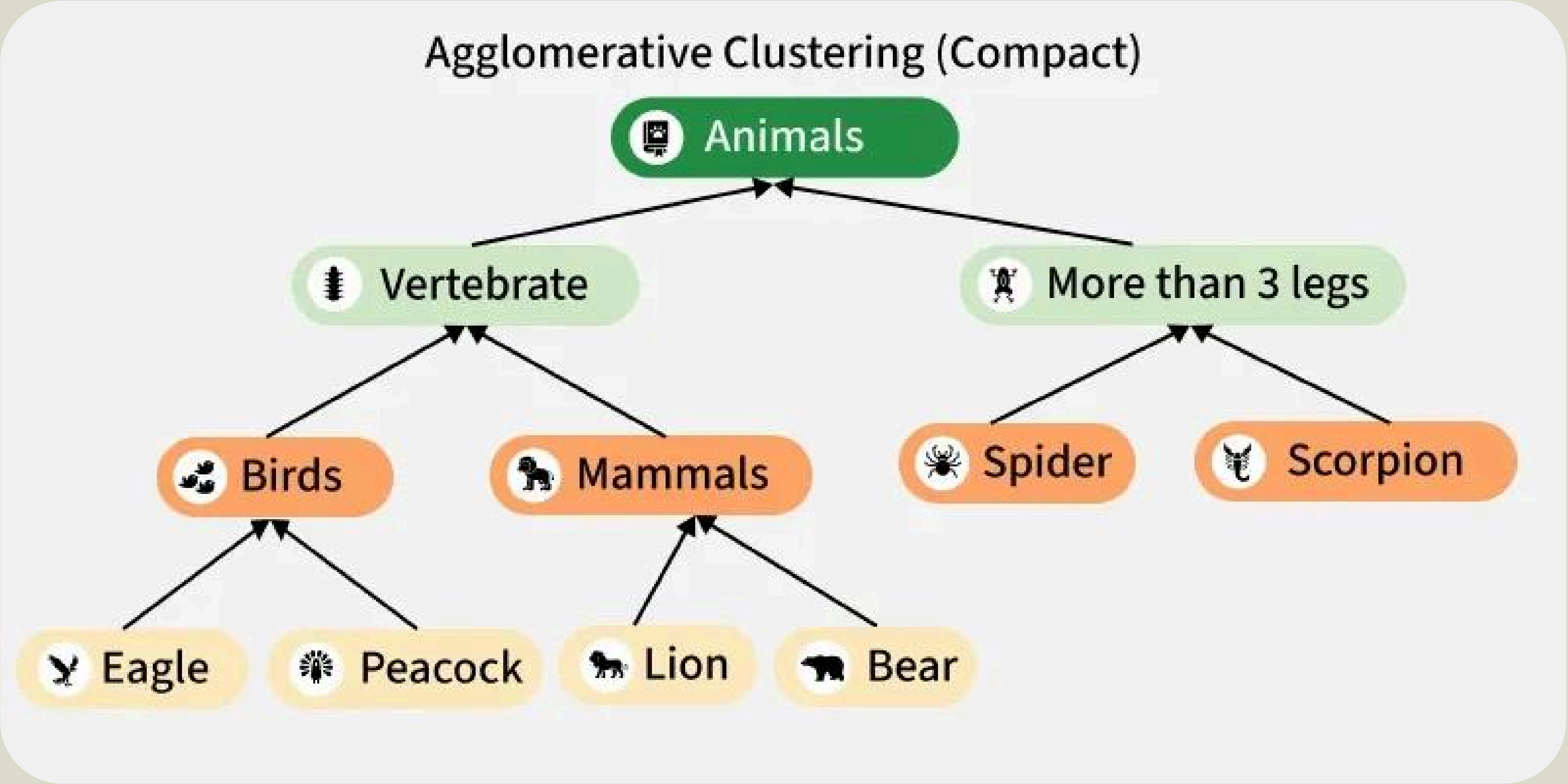
- Multi-objective evolutionary optimization engine
- Unsupervised Learning
- Inspired by natural selection principles
- Explore large design spaces
- Integrated in Grasshopper
- Evolutionary Iteration
  - Selection
  - Crossover
  - Mutation
- Optimizes conflicting objectives
- Outputs Pareto Front Solutions for informed decision making



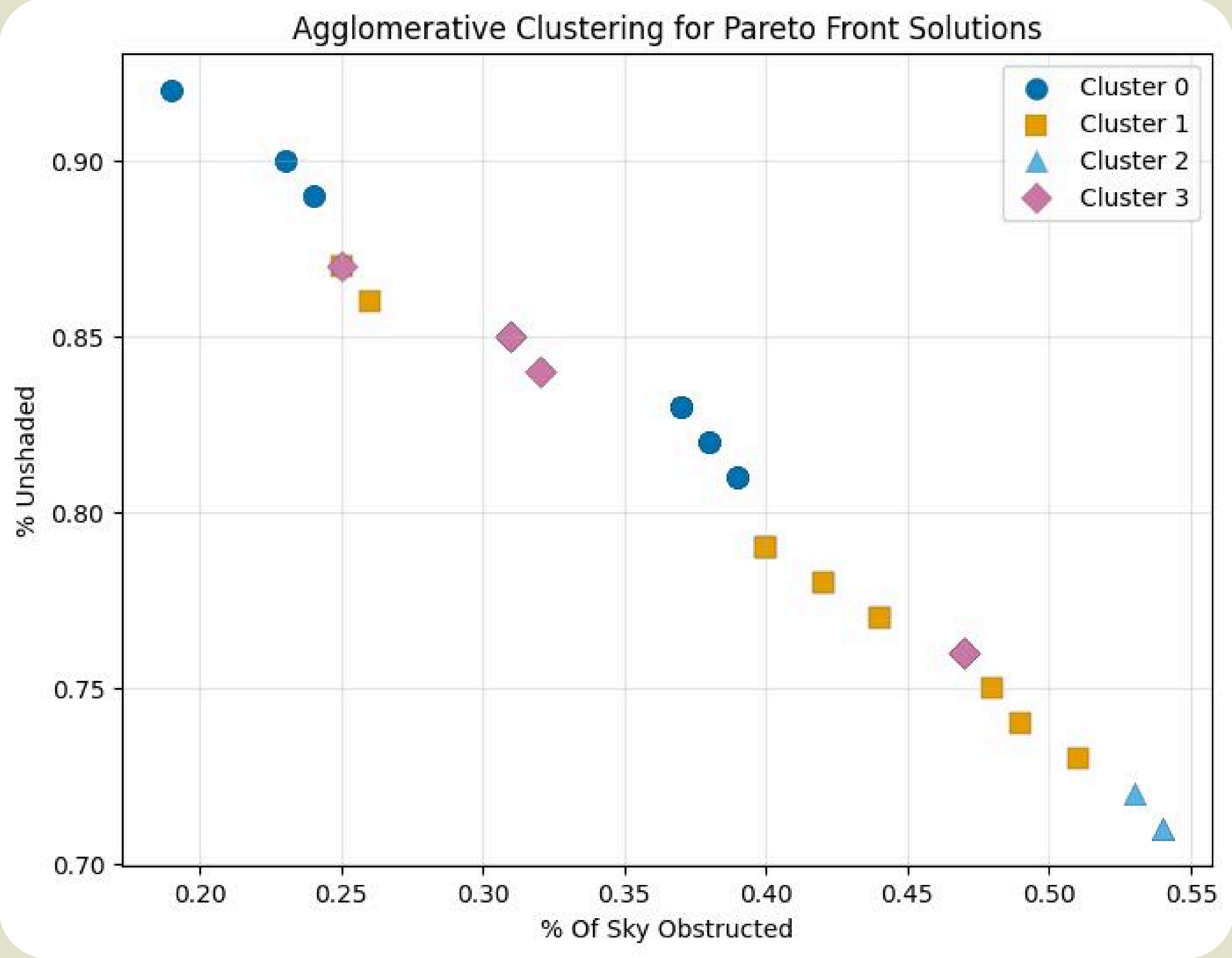
Classification Algorithm

Agglomerative Clustering

- Groups data points based on similarity of input features
- Unsupervised ML
- Creates a hierarchical structure of features
- Uses distance metrics to define similarity between data points
- Outputs a cluster labels for each solution
- Used to distinguish groups and reveal underlying structures in the data

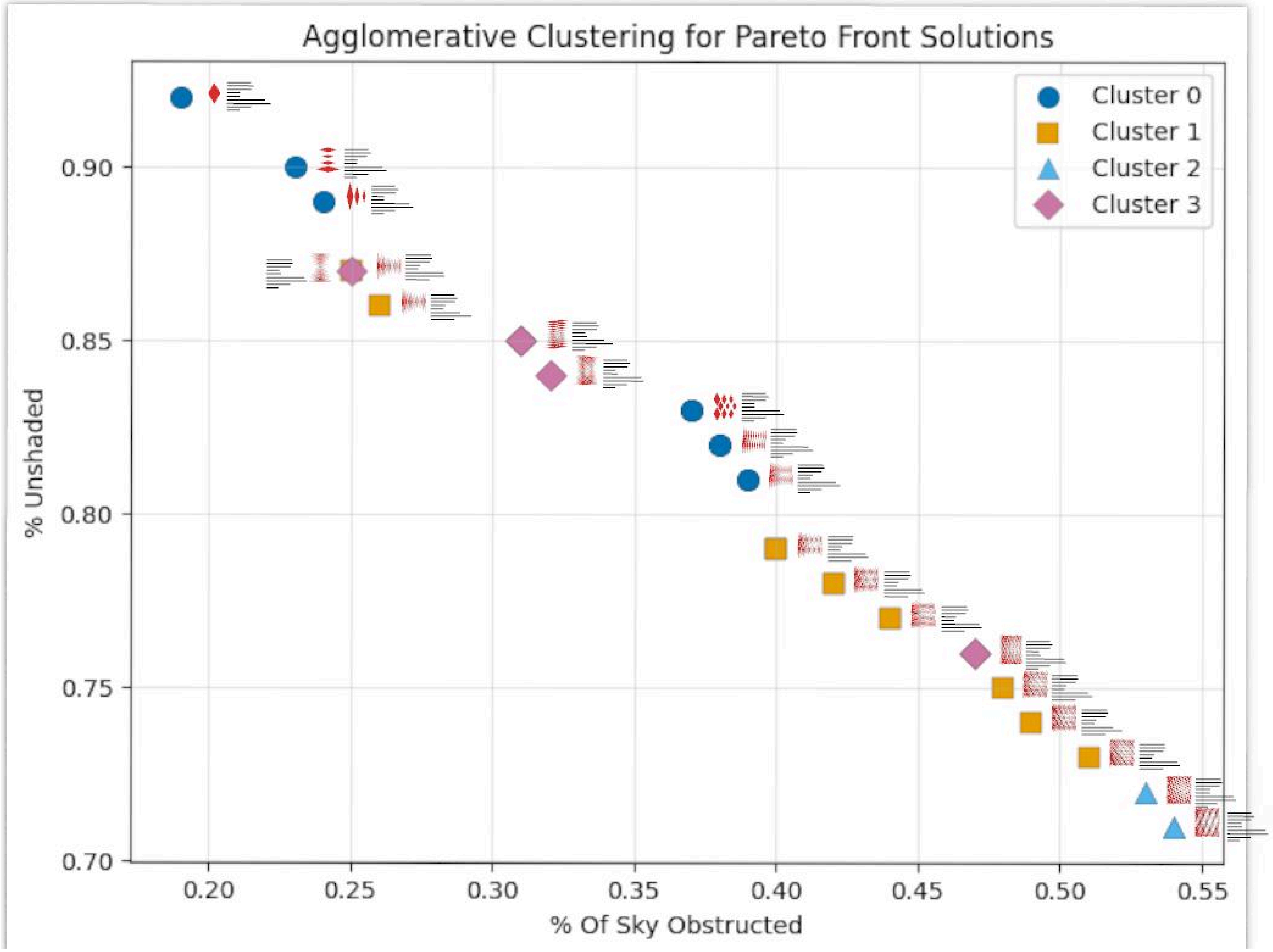


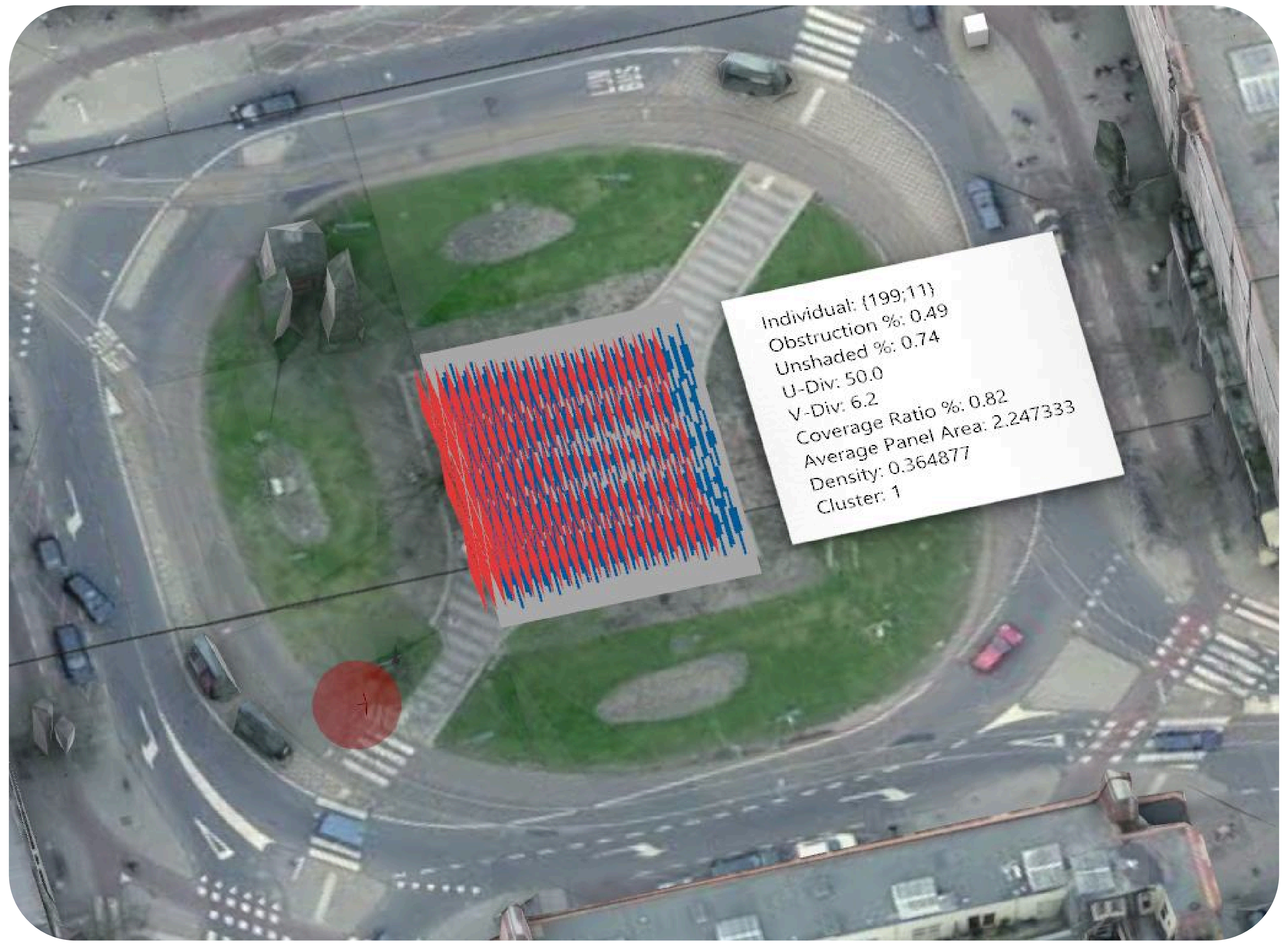
• ANIMAL CATEGORIZATION TREE - GEEKSFORGEEKS

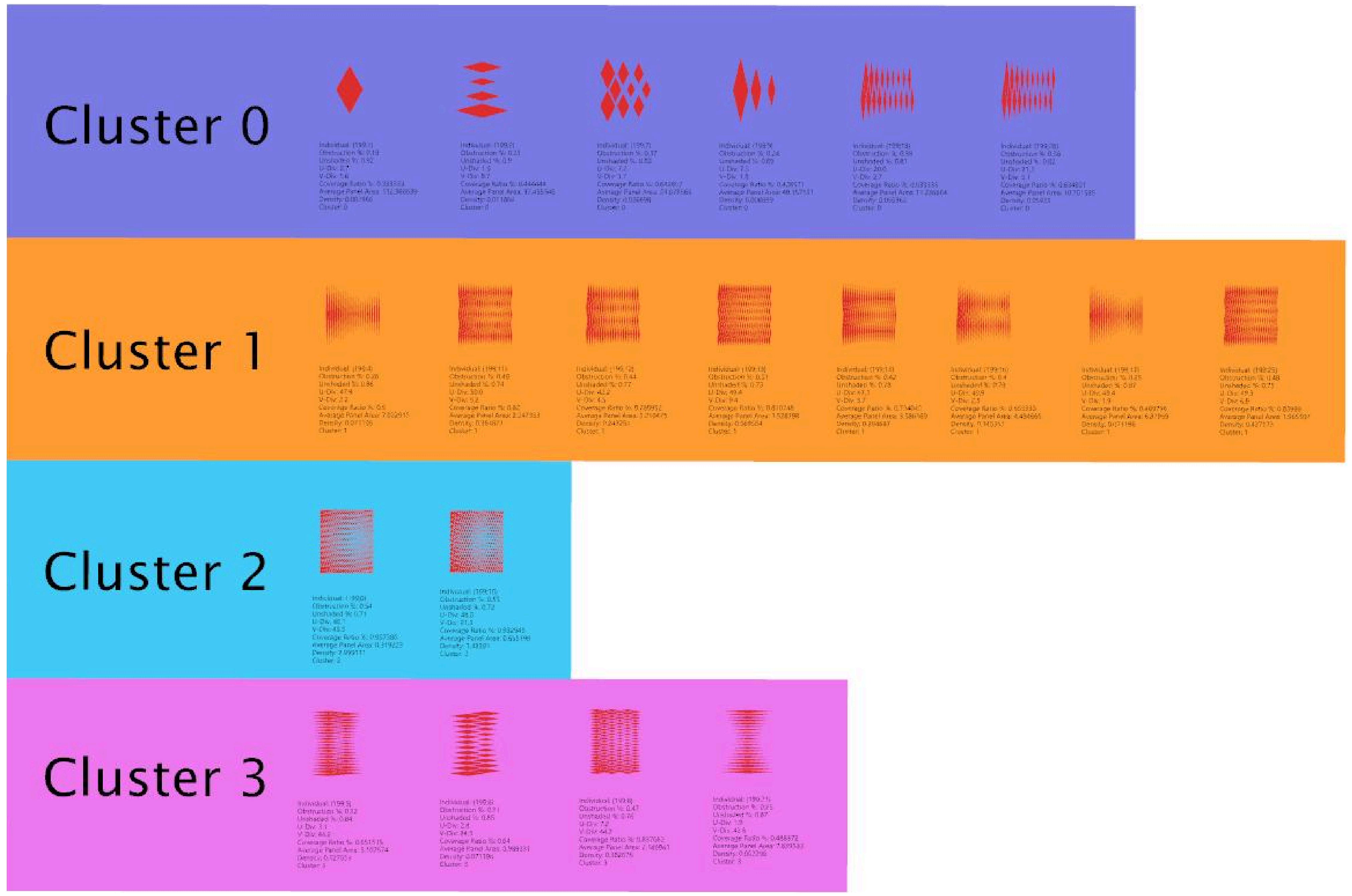


Presenting Designs and Choice

Displaying Pareto Front Solutions





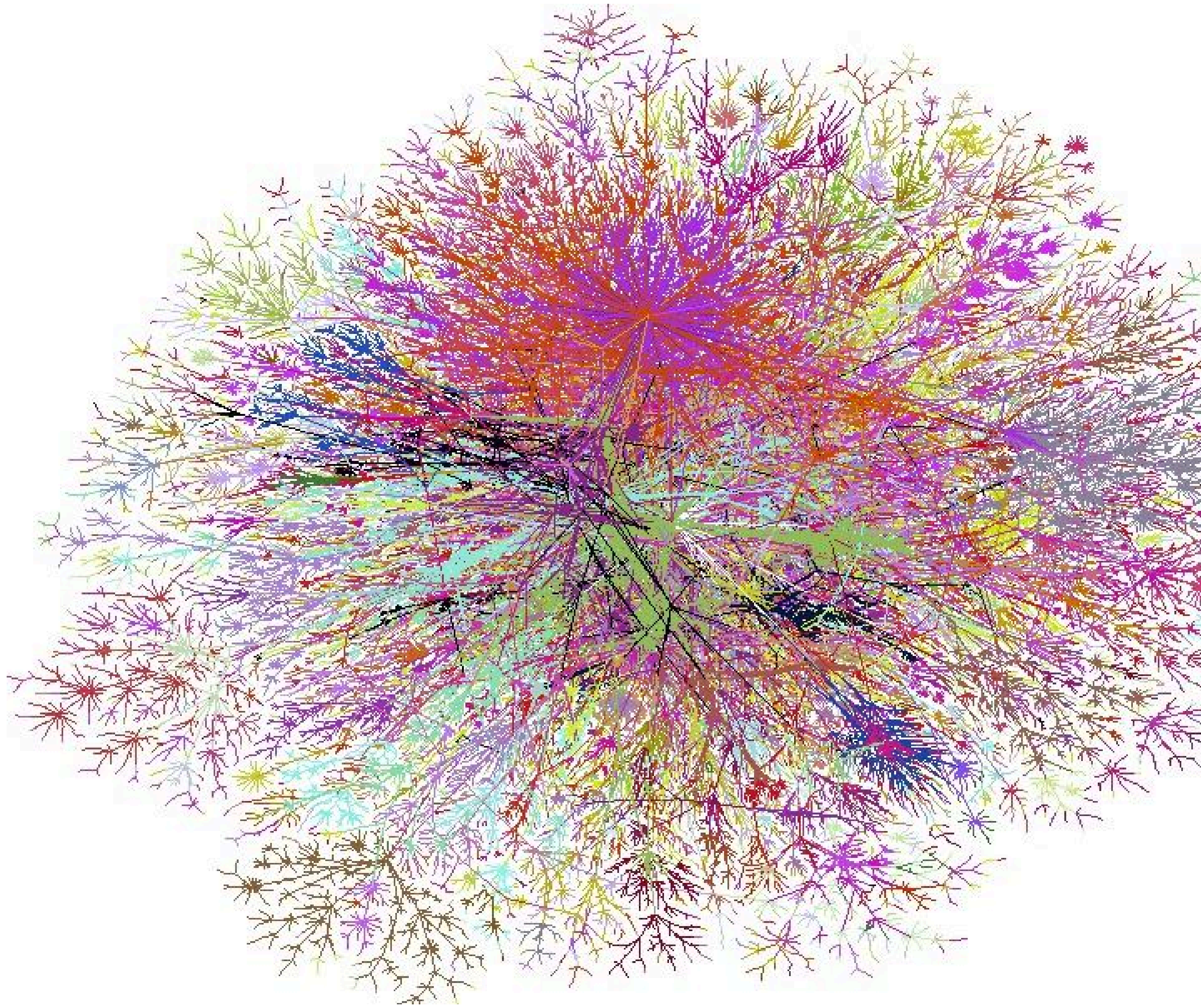


Limitations

- Evolutionary Algorithm was too greedy in it's approach
- Using shading instead of surface temperature
- Only one day of the year
- Only 2 gene inputs to WallaceiX
- Fixed canopy extrusion (10cm)
- Fixed canopy height (8.7m)
- Only one decimal point in U/V Div values
- Grid size for DirectSunHours too large

Computational Complexity

- Machine learning algorithms enable us to explore near infinite possibilities
- But we don’t yet have that kind of computational capacity
- Balance between accuracy and feasibility
- Acknowledging limitations



COMPLEXITY THEORY II (M. WOERMANN)

Implications for Designers ⟶ My Takeaways

- Algorithmic capabilities allow designers to shift from creating a single solutions to navigating entire solution spaces
- Machine learning can be used to extend people’s analytical capacities rather than replacing them
- Multi-objective optimization enables informed human-in-the-loop decision making
- Optimization is only as good as assumptions and inputs
- Emphasizing the importance of understanding limitations

AI Statement

Generative AI tools were utilized to support my work throughout this project. ChatGPT was used for assistance with coding tasks and to enhance the visual on slide 1. Additionally, the Raven AI plugin for Grasshopper was used to perform simple operations within the parametric design process; it is unfortunately not yet smart enough for more complex tasks. All design decisions, methodologies, and outcomes came from my own brain.

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